JS Math Object

1. **What is a Math Object in JS?**
    a. The Math object in JavaScript is a built-in object that provides a set of *mathematical constants*, *properties*, and *methods* to perform various mathematical calculations.
    b. *Math is a property of a Window Object*.
    c. Math object is static.
    d. It is not a constructor, so you cannot create instances of it.
    e. All methods and properties can be used without creating a math object first.
    f. **For Example**:
```
console.log(Math);  //Math {abs: ƒ, acos: ƒ, acosh: ƒ, asin: ƒ, asinh: ƒ, …}
console.log(typeof Math); // object
```
    g. **Syntax**:

> Math.propertyName;
>
> OR
>
> Math.methodName();

2. **Why Do We Need the Math Object in JavaScript?**
    a. In JavaScript, basic mathematical operations like addition (+), subtraction (-), multiplication (*), and division (/) can be done directly.

b. However, for more *complex mathematical tasks* like finding *square roots*, generating *random numbers*, or working with *trigonometric functions*, you need specialized tools.

3. **Properties of the Math Object:**

   a. **Math.PI:**

      i. The JavaScript *Math.PI* property is represents the mathematical constant π (pi), the ratio of a circumference of a circle to its diameter.

      ii. **Return Value**: A number approximately equal to 3.14159.

      iii. **Syntax**: `Math.PI`

      iv. **For Example** →

```
console.log(Math.PI); //3.141592653589793
```

   b. **Math.E:**

      i. The *Math.E* property in JavaScript is a static data property that represents the *Euler's number*.

      ii. **Return Value**: A number approximately equal to 2.71828 .

      iii. **Syntax**: `Math.E`

      iv. **For Example** →

```
console.log(Math.E); //2.718281828459045
```

4. Methods of the Math Object:

a. Here are some essential methods of the Math object, categorized for better understanding:

   i. Math.round()
   ii. Math.ceil()
   iii. Math.floor()
   iv. Math.trunc()
   v. Math.random():
   vi. Math.pow(base, exponent)
   vii. Math.sqrt()
   viii. Math.cbrt()
   ix. Math.max()
   x. Math.min()
   xi. Math.abs()
   xii. Math.sign()

1. Math.round():

a. The JavaScript *Math.round()* method accepts a numeric value as a parameter, rounds it to the nearest integer and returns the result.

b. **Syntax**: `Math.round(number);`

c. **Return Value**:

   i. If the decimal is >= 0.5 , it rounds up.
   ii. Otherwise, it rounds down.

d. **For Example** →

```
console.log(Math.round(4.3)); // 4
console.log(Math.round(4.7)); // 5
```

2. Math.ceil():

a. The JavaScript *Math.ceil()* method accepts a numeric value as a parameter and rounds UP to the smallest integer greater than or equal to that number.

b. If we pass an empty number or non-numeric value as an argument to this method, it returns " NaN " as result.

c. **Syntax**: `Math.ceil(number);`

d. **Return Value**: It will return the smallest integer greater than or equal to the number.

e. **For Example →**

```
console.log(Math.ceil(4.1)); // 5
console.log(Math.ceil(-4.1)); // -4
```

3. Math.floor():

a. The JavaScript *Math.floor()* method accepts a numeric value as a parameter and *rounds DOWN* to the largest integer less than or equal to the provided number.

b. **Syntax**: `Math.floor(number);`

c. **Return Value**: It will return the largest integer less than or equal to the number.

d. **For Example →**

```
console.log(Math.floor(4.9)); // 4
console.log(Math.floor(-4.9)); // -5
```

4. Math.trunc():

a. The JavaScript *Math.trunc()* method accepts a numeric value as an argument and returns the integer part of the number by truncating the decimal digits.

b. If the argument is 0 or -0, the result will be 0.

c. **Syntax:** `Math.trunc(number);`

d. **Return Value**: It will return the integer part of the number.

e. **For Example** →

```
console.log(Math.trunc(4.9)); //4
console.log(Math.trunc(-4.9)); //-4
```

5. Math.random():

a. The *Math.random()* method in JavaScript is used to generate a random decimal number between 0 (*inclusive*) and 1 (*exclusive*).

b. **Syntax**: `Math.random();`

c. **Return Value**: It will return the decimal number between 0 and 1.

d. **For Example** →

```
console.log(Math.random()); // It will generate a random number for each and every refresh.
```

6. Math.pow(base, exponent):

a. The JavaScript *Math.pow()* method is used to calculate the power of a base number raised to an exponent.

b. This method takes two arguments:

   i. Base

   ii. Exponent

c. **Return Value**: It will return the result of *base ^ exponent*.

d. **Syntax**: `Math.round(base, exponent);`

e. **For Example** →

```
console.log(Math.pow(2, 2)); // 4
```

7. **Math.sqrt():**
   a. The *Math.sqrt()* method in JavaScript accepts a numeric value as its argument and returns the square root of that number (non-negative number).
   b. **Syntax**: `Math.sqrt(number);`
   c. **Return Value**: It will return a non-negative number representing the square root.
   d. **For Example →**
```
console.log(Math.sqrt(9)); //3
```

8. **Math.cbrt():**
   a. The JavaScript *Math.cbrt()* method accepts a number as a parameter and calculates the cube root of a provided number.
   b. **Syntax**: `Math.cbrt(number);`
   c. **Return Value**: It will return a number representing the cube root.
   d. **For Example →**
```
console.log(Math.cbrt(27)); // 3
```

9. **Math.max():**
   a. The *Math.max()* method in JavaScript accepts a set of arguments as parameters, finds the *highest numeric value* among it, and returns it.
   b. **Syntax**: `Math.max( value1, value2, ..., valueN );`
   c. **Return Value**: It will return the maximum value among the inputs.
   d. **For Example →**
```
console.log(Math.max(10, 20, 30)); // 30
```

10. **Math.min():**

a. In JavaScript, the *Math.min()* method accepts any number of arguments, and it returns the minimum value among those arguments.

b. **Syntax**: `Math.min( value1, value2, ..., valueN );`

c. **Return Value**: It will return the minimum value among the inputs.

d. **For Example →**

```
console.log(Math.min(10, 20, 30)); // 10
```

11. **Math.abs():**

a. The JavaScript *Math.abs()* method accepts a number as a parameter and returns the absolute value of the provided number.

b. **Syntax**: `Math.abs(number);`

c. **Return Value**: It will return the non-negative value of the number.

d. **For Example →**

```
console.log(Math.abs(-12)); // 12
```

12. **Math.sign():**

a. The JavaScript *Math.sign()* method accepts a numeric value as a parameter, determines the sign of the specified number and returns the result.

b. **Syntax**: `Math.sign(number);`

c. **Return Value**:

i. 1 if the number is positive. 1 if the number is negative. 0 if the number is 0.

d. **For Example →**

```
console.log(Math.sign(10)); // 1
console.log(Math.sign(-15)); // -1
console.log(Math.sign(0)); // 0
console.log(Math.sign("text")); // NaN
```